

DUBOIS, PA, USA

WMO: 725125

Lat: 41.179N

Lon: 78.893W

Elev: 550

StdP: 94.89

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04787

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-14.7	-22.1	0.6	-16.0	-19.4	0.7	-13.5	11.8	-2.0	10.9	-3.7	4.2	260	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	29.3	21.1	28.0	20.4	26.9	19.8	22.7	27.1	21.8	25.9	21.1	24.9	4.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	17.1	25.0	20.6	16.3	24.1	19.8	15.5	23.3	69.5	27.3	66.4	25.9	63.6	24.9	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.7	7.9	DB	-20.5	31.5	3.8	2.2	-23.2	33.1	-25.5	34.4	-27.6	35.7	-30.4	37.3	(4)
(5)			WB	-20.8	24.3	3.4	0.8	-23.2	24.8	-25.2	25.3	-27.2	25.7	-29.7	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-4.2	-2.9	1.8	8.5	14.1	18.4	20.6	19.9	16.3	10.2	4.0	-1.1	(6)	
	DBStd	10.06	6.72	6.19	6.27	5.42	4.72	3.46	2.81	2.79	3.87	4.80	5.35	5.48	(7)	
	HDD10.0	1773	442	362	266	90	15	1	0	0	2	58	191	347	(8)	
	HDD18.3	3684	698	594	513	296	146	41	11	17	83	254	429	603	(9)	
	CDD10.0	1364	2	2	12	46	143	254	329	306	192	65	12	2	(10)	
	CDD18.3	234	0	0	1	2	16	44	81	64	23	3	0	0	(11)	
	CDH23.3	1552	0	0	2	23	132	289	546	393	153	15	0	0	(12)	
	CDH26.7	297	0	0	0	2	18	49	120	77	29	2	0	0	(13)	
(14) Wind	WSAvg	3.6	4.6	4.4	4.3	4.2	3.5	3.0	2.7	2.5	2.8	3.5	4.0	4.4	(14)	
(15) Precipitation	PrecAvg	1087	69	66	93	83	98	115	99	101	96	71	86	89	(15)	
	PrecMax	1435	165	139	170	123	231	373	240	178	151	131	222	178	(16)	
	PrecMin	883	27	23	32	30	23	17	47	17	18	18	33	32	(17)	
	PrecStd	112	32	31	35	29	49	75	48	41	35	31	40	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	16.3	22.0	26.2	28.7	29.8	31.3	30.6	29.6	25.2	19.8	16.1	(19)	
		MCWB	12.0	10.4	13.6	15.9	20.0	21.4	22.1	21.2	21.0	18.2	12.7	12.8	(20)	
	2%	DB	12.4	12.0	17.9	23.0	26.7	28.0	29.3	28.7	27.2	22.4	17.1	12.6	(21)	
		MCWB	10.5	8.6	11.0	13.8	17.9	20.7	21.6	21.0	19.6	16.2	12.1	10.6	(22)	
	5%	DB	8.7	8.6	14.5	20.2	24.7	26.7	27.8	27.2	24.9	20.3	14.7	9.6	(23)	
		MCWB	6.9	5.3	9.4	12.3	17.0	19.8	20.8	20.1	18.6	15.4	10.7	7.3	(24)	
	10%	DB	5.2	6.1	11.7	17.6	22.4	24.9	26.5	25.7	22.9	17.9	12.6	6.6	(25)	
		MCWB	3.5	3.3	7.8	10.8	16.0	18.7	20.0	19.4	17.7	14.0	9.5	4.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	12.2	14.6	17.4	21.2	22.8	24.1	23.5	22.6	20.2	15.3	13.5	(27)	
		MCDB	14.5	15.2	20.0	23.4	26.4	27.6	29.0	27.9	27.1	23.6	18.7	15.0	(28)	
	2%	WB	10.6	8.9	12.5	15.4	19.7	21.8	23.0	22.3	20.9	18.1	13.1	10.8	(29)	
		MCDB	12.3	11.6	16.1	20.3	24.7	26.3	27.6	26.3	24.7	20.6	15.6	12.0	(30)	
	5%	WB	7.0	5.7	10.5	13.9	18.5	20.9	22.0	21.4	19.9	16.3	11.6	7.8	(31)	
		MCDB	8.3	8.0	14.1	18.5	22.8	25.1	26.0	25.0	23.4	19.1	14.0	9.2	(32)	
	10%	WB	3.4	3.3	7.7	12.2	17.2	20.0	21.2	20.6	18.8	14.5	9.7	4.8	(33)	
		MCDB	5.0	5.7	11.4	16.4	21.2	23.6	24.8	23.9	21.8	17.5	12.6	6.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.2	9.3	11.1	10.9	10.2	10.4	10.3	10.2	9.4	7.9	6.5	(35)	
		MCDBR	9.6	11.5	13.2	14.3	12.9	11.8	11.5	11.8	12.2	12.0	11.1	10.3	(36)	
	5% WB	MCWBR	8.2	9.0	8.3	7.9	6.4	6.0	5.3	5.4	5.9	6.7	7.7	8.7	(37)	
		MCWBR	9.3	10.7	11.8	11.8	10.8	10.0	9.9	9.6	9.9	10.0	9.8	9.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.300	0.322	0.342	0.363	0.398	0.437	0.441	0.431	0.366	0.355	0.315	0.295	(40)	
		taud	2.410	2.326	2.370	2.367	2.281	2.193	2.196	2.217	2.383	2.413	2.506	2.484	(41)	
	Ebn at Noon	Ebn at Noon	855	883	907	908	879	841	834	831	870	836	835	835	(42)	
		Edh at Noon	80	101	109	118	131	144	142	135	107	93	73	69	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.47	2.28	3.28	4.43	5.10	5.57	5.67	5.08	4.11	2.70	1.79	1.18	(44)	
	RadStd	RadStd	0.12	0.29	0.34	0.46	0.50	0.48	0.42	0.30	0.47	0.29	0.24	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+102	N/A

Nomenclature: See separate page